# BioSpecInfo

## Traceability Matrix — BioSpecInfo

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Versione **bsi-v216** · commit `e2b16f3`

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# Traceability Matrix — BioSpecInfo

| Field | Value |
|---|---|
| **Software** | BioSpecInfo |
| **Author** | Samuele Pio Provenzano |
| **Version described** | `bsi-v216` |
| **Purpose** | Connect every declared requirement to the implementation that realises it and to the bench that verifies it. |

> **How to read this matrix.** Every row is a closed chain: a requirement, the point in the code that realises it, and the bench that verifies it works. A row without a bench is an **unverified** requirement and is declared as such, not hidden: the usefulness of a matrix lies precisely in making the gaps visible.
>
> The outcomes of the benches cited here are reported in full, with their complete output, in `../evidence/RAPPORTO-VERIFICA.md`, which is generated automatically.

## 1. Scientific requirements

| ID | Requirement | Implementation | Verification bench |
|---|---|---|---|
| **SCI-01** | A drug structure must come from an external source and match that source, not merely weigh correctly | `FARM_DATA` in `index.html` — **263 entries**, 36 declaring a ChEMBL identifier and formula | `audit_farmaci` — the formula is **rebuilt by counting the graph's atoms** and compared with ChEMBL's: 36 of 36. Weight alone was not enough: three pre-existing entries (lenalidomide, palbociclib, aripiprazole) were **isomers** — same weight, different molecule — and were replaced. Two ratchets: verified provenance ≥ 36, readable structures ≥ 240 |
| **SCI-02** | Functional groups must be recognised on the structure, not on the text of the SMILES | `bsi-spettri.js` §1, SMARTS patterns via RDKit | `test_spettri` — 8 reference molecules, additive recognition |

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|---|---|---|---|
| **SCI-03** | A molecule with several functional groups must show them all | `bandeDaGruppi()`, additive construction | `test_spettri` §1 — acetylsalicylic acid: ester **and** acid |
| **SCI-04** | Spectra must respect the conventions of their technique | `svgIR()`, `makeNMRsvg()`, `drawUVSpectrum()`, `drawMSSpectrum()` | `test_assi` (SVG), `test_assi_canvas` (canvas) |
| **SCI-05** | A computed spectrum must be reproducible | every `Math.random()` removed from the traces | `test_assi` §3, `test_spettri` §8 — byte-for-byte comparison |
| **SCI-06** | The height of ¹H peaks must represent the integration, or not be asserted at all | `makeNMRsvg()`; `_nmrSvg()` draws bands of uniform height | `test_assi` §4 |
| **SCI-07** | Conjugation must act on the individual carbonyl | SMARTS `acidoConiug`, `estereConiug`, `chetoneConiug`, `aldeideConiug` | `test_spettri` §1 — aspirin 1760/1690; acetophenone 1690 |
| **SCI-08** | An ambiguous physical constant must not be resolved arbitrarily | `costante_fisica` in `bsi-ai-hub.js` — collects every candidate | `test_costanti` — 45 checks, including the refusals for ambiguity |
| **SCI-09** | The limits of the predictors must be declared to the user | chart headers; `docs/06` §3.5 | `verifica_guida` — consistency between promises and code |
| **SCI-10** | Molecular symmetry must determine the selection rules correctly | `ssimm` section in `index.html` | `test_simmetria` — 26 checks: normal modes, centre of inversion, mutual exclusion |
| **SCI-12** | A chromatogram must follow from the parameters, not be drawn by hand | `scroma` section: σ = t<sub>R</sub>/√N, R<sub>s</sub> from distance and widths | `audit_stabilita` (opening and drawing), inspection of the values against the definitions |
| **SCI-13** | A calibration line must declare the uncertainty of its coefficients and recognise the two ways it misleads | `staratura` section: ordinary least squares, F-test line⟷parabola, deleted residual | Verified on the three example sets included: linear (F = 0.3), non-linear (F = 186.6 > 7.71), with an outlier (1 detected) |

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|---|---|---|---|
| **SCI-14** | A QSAR model must be evaluated on scaffolds never seen, not on analogues from the same set | `bsi-cheminfo.js` §9 — `divisionePerScaffold()` alongside `divisioneCasuale()` | `test_cheminfo` — the scaffold split leaves no scaffold shared between training and test |
| **SCI-15** | A model score must be compared against chance, not presented on its own | `bsi-cheminfo.js` §9 — retraining on scrambled labels, eight repetitions | `test_cheminfo` — verified **in both directions**: on a learnable signal $R^2$ 0.394 beats all eight twins (margin 0.287); on random labels $R^2$ −0.199 does **not** beat them |
| **SCI-16** | Two spellings of the same molecule must not be counted twice, nor land on opposite sides of the split | `bsi-cheminfo.js` §1 — deduplication on the canonical SMILES after the largest fragment | `test_cheminfo` — `OC(=O)C` and `CC(=O)O` collapse into one entry; salts lose the counterion |
| **SCI-17** | Similarity between molecules must be computed, not estimated | `bsi-cheminfo.js` §3 — Tanimoto and Dice over Morgan, RDKit, pattern and MACCS fingerprints | `test_cheminfo` — compared against hand-computable values, including the 0/0 = 1 convention |
| **SCI-18** | A validation on a single split, over a small set, is a noisy number: it must come with its dispersion | `bsi-cheminfo.js` §9-bis — `validazioneIncrociata()` over **scaffold-grouped** folds | `test_cheminfo` — the folds cover every molecule once, **no scaffold sits in two folds**, and the same seed gives the same folds |
| **SCI-19** | A result that changes on every run is verifiable by nobody | seeded mulberry32 generator for splits, scrambles and folds; the seed is written into the report | `test_cheminfo` — two calls with the same seed give identical folds; the method report declares the seed |
| **SCI-20** | Results must be able to leave the tool, and the method must leave with them | `bsi-cheminfo.js` §9-ter — `esportaCsv()` and `rapportoMetodo()`, both formed in memory in the browser | `test_cheminfo` — columns consistent with the header, a name containing a comma quoted, residual computed, empty cell for missing activity; the report carries seed, fingerprint, verdict, limits |

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|---|---|---|---|
| **SCI-21** | A number presented as structural similarity must come from a fingerprint, and one implementation must compute it | `rdkit_lab.html` loads `bsi-cheminfo.js` and uses its `tanimoto()` and `fingerprint()`; the eight-bit similarity has been removed | `test_cheminfo` — page and engine give the same Tanimoto over all **28 pairs**, deviation **exactly 0**; the fake-similarity functions no longer exist; every reference structure is valid and carries the **monoisotopic mass** of the drug it names |
| **SCI-22** | Conversions between molecular languages must be checked against an external source or against hand calculation, and coordinates written to a file must be real and declared | `bsi-molingue.js` — **seventeen outputs**, from SMILES to CML | `test_lingue` — 7 **literature InChI keys**, full SMILES → molfile → SMILES round-trip, 26 hand-written names, **7 required refusals**; mass composition against hand calculation (alanine C 40.44%, aspirin C 60.00%); XYZ and PDB with non-null coordinates **and** declared 2D |
| **SCI-23** | Angles, lengths and dihedrals are measured on the coordinates, and on flat coordinates they are not shown | dot product for angles, cross product for dihedrals | `test_mol3d` — tetrahedron **109.4712°**, water 104.47°, $CO_2$ 180°, $BF_3$ 120°, ammonia 106.13°; dihedrals 0°, 60°, 90°, 180°; ethanol C–C 1.509 Å; and the refusal tested in both directions |
| **SCI-24** | A spectrum read from a file must be decoded according to the format, and the peaks found must be the ones that are there — no more, no fewer | `bsi-spettrolettore.js` — JCAMP-DX with full ASDF (PMAI, DIF, DUP) and the integrity check the format prescribes; peaks by **prominence**, noise from MAD × 1.4826 | `test_spettrolettore` — **61 checks**. On three Gaussians built at 1715, 2950, 3400 $cm^{-1}$ under noise it finds three: 1716, 2952, 3402. **Both ways**: on pure noise it finds **zero**. ASDF verified symbol by symbol (`abcdefghi` = −1…−9; `n` = −5, not −4) |
| **SCI-25** | A band must be declared *compatible with*, never assigned to a single possibility | `BANDE_IR` — 24 ranges; `assegnaIR()` returns **all** compatible assignments | `test_spettrolettore` — at 1715 $cm^{-1}$ both a ketone C=O and a carboxylic acid C=O appear; the section states that it does not deduce the structure |

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| **SCI-26** | A published model must be checked against **its own** declared error, and a learned model against chance | `bsi-cheminfo.js` §10 — ESOL (Delaney 2004) with its uncertainty, four drug-likeness filters reported separately, random forest with out-of-bag error | `test_spettrolettore` §4-5 — ESOL against four experimental values, mean deviation **0.68**, inside the declared ~1 log unit; forest out-of-bag $R^2$ **0.90** on a non-linear relationship and **−0.16** on pure noise; deterministic for a given seed and different for another |
| **SCI-27** | A structure without a source must not get in, not even one that is known | collected from ChEMBL, `tools/dati/farmaci_v187.json` carries each entry's identifier | **Two refusals**: ivermectin (`structure_type NONE`, a mixture of homologues) and semaglutide (a protein, `SEQ`). And two entries kept under the **record's** name rather than the query's: «morphine» → apomorphine, «levothyroxine» → liothyronine |
| **SCI-28** | A predicted spectrum must say WHICH atoms produce each signal, and the deviation must be measured on molecules that did not choose the parameters | `bsi-pretsch.js` — the estimation tables transcribed in full from the source (Pretsch 4th ed.: 91 $^{13}C$ and 66 $^{1}H$ rows for benzenes, 42 for ethylenes, 31 for alkanes, 24 for $^{13}C$ aliphatics, 4×4 steric corrections); `bsi-nmr.js` — the reasoning that applies them: ring position obtained by walking the cycle, additive scheme with a cyclic reference compound, chemical equivalence by shell environment code | `test_nmr` — **56 checks**. Two separate sets: tuning **0.71 ppm** over 9 molecules, **validation 0.92 ppm** over 22 never used for tuning, worst case **4.9 ppm**, $^{1}H$ **0.06 ppm** (0.03 on the aromatics alone). The bench requires validation to stay **worse** than tuning: were they equal, someone would have moved a molecule between the two sets |
| **SCI-29** | Chemically equivalent atoms must give ONE signal, not one per atom | environment code built as concentric shells by distance (the HOSE-code principle), with aromatic bonds written as such and not in Kekulé form | `test_nmr` — eight molecules whose signal count is known: benzene **1** (not six), toluene 5, p-xylene 3, naphthalene 3, biphenyl 4, aspirin 9. All correct |

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| **SCI-30** | From an image one can recover the SHAPE of a spectrum, not its calibration | `bsi-digitalizza.js` — `bsi-spettrolettore.js` delegates `daImmagine()` to it, a single implementation. The axis frame is found first, the trace is extracted at **sub-pixel** resolution (darkness-weighted centroid), the run is chosen by continuity **and** by darkness, both axes are calibrated linearly or logarithmically, %T is converted to absorbance with $A = -\log_{10} T$ | `test_digitalizza` — **45 checks** over seven constructed figures: mean deviation **2.2 cm⁻¹**, worst **7.3** (one column is worth ~4 cm⁻¹). **Both ways**: from a blank sheet and from the app icon no curve comes out, and the crude fallback on the same figure with frame and labels finds **one** peak instead of three. `test_spettrolettore` for the routing |
| **SCI-31** | A two-dimensional NMR map must say WHICH atoms produce each correlation, and must not connect what the topology does not connect | `bsi-nmr2d.js` — one-bond HSQC with inverted sign on $CH_2$, three-bond COSY between non-equivalent protons, two- and three-bond HMBC; exchangeables excluded from off-diagonal spots | `test_nmr2d` — **40 checks**, in both directions: hexafluorobenzene gives no HSQC spots, benzene **no** off-diagonal COSY spots (its six protons are equivalent), methane **no** HMBC spots despite having four protons |
| **SCI-32** | A program-built 3D geometry must carry THE SAME indices as the predictor, otherwise the peak↔atom link lights up the wrong atom | `bsi-geom3d.js` — distance geometry over the `bsi-nmr.js` graph, hydrogens appended at the end, generator seeded from the SMILES | `test_geom3d` — **29 checks**: every heavy atom has the same index and element across five molecules; bonds and angles within tolerance of literature values; **in both directions** benzene comes out planar and cyclohexane does not |
| **SCI-33** | A document must be opened in full, and whatever could NOT be read must be declared — a reader that returns little without saying so makes the document look nearly empty | `bsi-documento.js` — PDF page by page with its text layer (PDF.js), images, text, Office archives opened with `DecompressionStream` and no libraries | `test_documento` — **47 checks**: PDF, Word and text genuinely opened and their content verified; **in both directions** an image declares there is no optical character recognition and produces no working, an unknown format is refused rather than guessed, a binary file named `.txt` is recognised as binary |

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| **SCI-34** | Spectroscopic data recognised in a text must carry their PROVENANCE, and the working must declare what it is missing | `bsi-quesito.js` — recognition by section (IR, MS, $^{1}H$, $^{13}C$) with the boundaries between sections, then `bsi-elucida.js` for the step-by-step working | `test_documento` — formula, bands, masses with intensities, $^{1}H$ signals with integral and multiplicity, $^{13}C$ signals; **in both directions**: $CDCl_3$ does not become the compound, a band outside 400-4000 is not a band, without an MS label there are no masses, a new paragraph closes the section, a text with no spectra produces no data |
| **SCI-35** | A figure with a frame, labels and a grid must not be digitised as if it were only the curve: the axes and the numbers are dark pixels just like the trace | `bsi-digitalizza.js` — the frame is sought among the CONTINUOUS lines in the margins, and a border is told from a gridline by its **irregular spacing** relative to the family **or** by its thicker, darker stroke; the crop is moved past the line thickness; vertical gridline columns are marked as not measured and reconstructed from their neighbours | `test_digitalizza` §1-2 — on a figure with a frame **and** a grid (11 vertical families, 7 horizontal) it finds all four sides and the peaks stay within 2.2 $cm^{-1}$; **both ways**, on a grid-only figure it does **not** invent a frame and says so, and a **flat baseline** is not mistaken for an axis |
| **SCI-36** | A digitised curve must be able to leave the application, and the exported file must be valid | `bsi-digitalizza.js` — CSV output with the units in the header, and JCAMP-DX 4.24 in (XY..XY) form with FIRSTX/LASTX/NPOINTS and the provenance written inside the file | `test_digitalizza` §4 — the **full round trip**: the exported JCAMP is read back by `bsi-spettrolettore.js`'s `leggi()` and returns with the same number of points, the same end points and the same units |

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| **UI-14** | One function must not have two entry points. The spectra lived in **nine sections** across two menus; Spectra in three places; and the ✨ panel showed **two "Data Science", two "File Manager", two "Articles", two "Updates"**, because the panel has a hardwired list that opened the very sections the menu module was inserting — and nobody compared the two lists | `bsi-centrospettri.js` — the panels are **moved** (`appendChild` moves, it does not copy) into a single tabbed section; the old buttons stay hidden because they are the target of `goSection()`, and a tab **clicks the button** instead of showing the panel, so lazy initialisation goes through the one road that already existed. `bsi-menu.js` — the "Utility" category is in the **main bar** with all its entries, and the rule is: if something has a **reachable** button in the bar it is not in the ✨ menu, if it has none it is only in the menu. "Reachable" is not measured with `offsetParent` — the bar shows one group at a time, and a perfectly reachable button has a null `offsetParent` | `test_articoli` §5-6 — the **ten** tabs are inside the centre, and each identifier exists in the document **exactly once** (the proof that it was moved and not copied); `goSection()` on each opens the centre on the right tab and shows **only** one. `test_menu` — Utility visible in the bar with all **fifteen** entries plus the two language ones, **no** menu entry repeating a bar section, **no** duplicate label in the panel, across **two viewports**: large screen and phone. **Both ways**: no repeated id, no duplicate button, Spectra and the reader out of the bar but with their button still in the document because `goSection()` uses it |
| **UI-15** | A data table must not exist twice: two copies diverge, and the reader gets a different answer depending on where they came in | The J coupling constants existed in two independent tables with **different** numbers (aromatic ortho $^{3}J$ 6–9 Hz against 7–10 Hz). The one inside the Spectra centre remains, which sits in the section with its own reference list; `_apriCostantiJ()` is the single road and leads there | `test_articoli` §6 — asking for the J constants opens the centre on the reference-tables tab, on the right sub-tab, and the table has its rows |

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| **ING-02** | A hand-written list of articles is stale the next day, and a prize year written from memory is an error nobody can see | `bsi-articoli.js` reads **live** from Crossref and from the Nobel Foundation; `bsi-articoli-nobel.js` holds **explanations only**, matched to the topic of the official motivation and not to the year, so no attribution comes from the application. The addresses are not written: they are **pure functions** of the identifier (DOI, year) | `test_articoli` §1-4 — the links tested as functions (stable for the same datum, different for a different one); the explanations matched against nine test motivations and **none** invented when the topic is not recognised; and the proof that counts: a response is **injected** with a name that exists nowhere, and the screen must show **that** name. **Three distinct failures tested**: failed request, empty answer, unrecognised shape — and they must be stated **differently** |
| **UI-16** | Changing language must cost a gesture, not a navigation | `bsi-sellingua.js` — a **round** button top right, at the title's height, with the flag in use; it opens into a disc with the fourteen flags on **two shells** (six inside, like a benzene ring) and, at the centre, the language name and **how much it is translated, measured** by `coperturaPerLingua()`. The actual switch stays in `bsi-lingue.js`: here there is only the way to call it, and the "Language" section remains because it INFORMS while this one ACTS | `test_atomic` §1 across **two viewports** — round, top, right, and "at the title's height" **measured** as the overlap of the two boxes (with `align-self:center` it lined up on a large screen and not on a phone, because there the flex aligns to the top and the logo block is two lines tall); an accessible name that says what it does rather than "flag of Italy"; the disc **inside the screen**; and **both ways**: the language really changes and going back to Italian the text is identical |

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|---|---|---|---|
| **SCI-37** | A didactic game's questions cannot be a second source of chemical truth, and cannot have two right answers | `bsi-gioco-domande.js` — it **contains no questions**: nineteen generators drawing from `ELEMENTS`, `SPEC_DATA`, `RXN`, `RETRO_STRATEGIES`, `AA_DATA` and the `BSIElucida` / `BSILettoreSpettri` engines, and every question carries its **provenance**. Range distractors are **disjoint** from the answer's, because different groups' ranges overlap ($^4$J meta 1–3 Hz against $^4$J allylic 0–3 Hz) | `test_atomic` §2 — the nineteen generators **each tried** (one was dead and "available" called it running, because it looked at the source instead of trying it); the range reader on six known cases (it read "1–3 Hz" as [−3, 1], and the consequence was not a skewed number: it was that the four range generators stopped coming out and the "zero ambiguous" check passed vacuously); 600 sound questions; **zero ambiguous**, with the guard that those generators do run |

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|---|---|---|---|
| **UI-17** | A game must be playable from start to finish — exploration, puzzles, synthesis, guardians — and must not consume when nobody is watching | `bsi-atomic.js` + `bsi-atomic-livelli.js` — **Atomic Protocol**: **a hundred levels** in ten zones, generated from a seed (the level number) on a grid where two neighbouring rooms share the wall and the door, so doors match by construction; **ten guardians** (one every ten levels, in arenas at least 14 m wide) whose phases ask for a compound, an element or an answer; twelve recipes with equations **balanced atom by atom**; generated puzzles; **thirty audio logs** (three per zone, seven voices) heard while playing, with the device's voice, subtitles and an archive in the Codex, piecing together the mystery of the command phrase; **saving** at every level; the loop stops when the section is not visible; moving between levels **frees the previous level's geometries and textures** | `test_atomic` §3-5b, §6, §8 — the thirty logs all placed in the levels; the hundred levels generated and walked (twin doors, no unreachable or overlapping room, bench-terminal-exit in each, the elements needed to beat every guardian present before it, ten distinct guardians, wide arenas), the equations counted and **a wrong equation rejected**; level 1 played up to the elevator and the save; the guardians **both ways**; PROTOCOL 12 at level 100 ends the game; GPU memory that **does not grow** across ten levels with four guardians; the loop stopping when the section closes; the game's **floating button** above Spectra, aligned, with an icon and accessible name, opening the game, and the floating buttons not covering the scene in game and coming back on exit |
| **UI-18** | A view, difficulty and graphics choice must stay chosen | `bsi-atomic.js` — **first person by default** (third as an option), four difficulties (Easy, Normal, Hard, Experiment — where the reaction must be balanced before synthesising it), **high, medium or PS1 graphics**, sound and language: all in the settings menu, also reachable from pause, and all saved to local storage, with every read guarded because private browsing may refuse to write. The starting graphics is high (PS2 level), «medium» on a phone with few cores; PS1 remains a retro option | `test_atomic` §6-7 — the chosen view, difficulty and graphics really end up in storage and are read back; there are four difficulties and **«Experiment» is not «Easy» renamed**; changing graphics mid-game **does not lose the game** |

| ID | Requirement | Implementation | Verification bench |
|---|---|---|---|
| **UI-19** | A declared graphics quality — the PlayStation 2 level — must be built, not evoked, and without horror | `bsi-atomic-figure.js` + `bsi-atomic-grafica.js` — the protagonist with **thousands of polygons**: a dense sphere head shaped by hand (skull, brows, cheekbones, nose with wings, lips, chin) with **modelled eyes** (eyeball, iris, eyelid) and a face painted with soft shading and no outlines, a hair cap with strands, **five-fingered hands** in nitrile gloves, a lab coat with modelled lapels, collar, pockets, buttons and badge, shoes from an extruded profile; high-resolution textures with **normal maps**; **glow** from additive sprites on lamps, screens and cores; the ten guardians are **the institute's machines** (pump with hydraulic arms, extruder, laser bench, sequencing robot, electrolysis, cryostat, automated pharmacy, hot cell, clarifier, reactor), no creatures; residues are solvent spills. The **environments are art deco**, in the spirit of BioShock: checkered marble with brass inlays, walnut wainscot, green and gold wallpaper, marble pilasters, fan sconces, brass-ringed ceiling lights, coffered ceilings, Institute and Albedo Consortium posters, chalk writing on the walls, drips falling into puddles, amber light. High graphics is the default; PS1 remains a retro option. **The game is played in first person**, with an art deco HUD (segmented health and reagent in brass frames): the arms (from models supplied by the user, in `models/` and prepared by `tools/prepara-modelli.js`: **real skinned hands** in a nitrile glove whose poses — the grip, the finger on the trigger, the open palm, the idle gesture — are frames of their own animation blended finger by finger; the **lab-coat sleeve** cropped from a scan, with the light-blue shirt cuff; the **capsule gun** in the right hand, fingers closed on the grip, the capsule in the compound's colour and the spray from the muzzle; until the models arrive the procedural arms stay; each finger follows its pose on a **damped spring**, with the micro-motions of a living hand, the trigger pulled sharply, the recoil in the wrist, the hand opening before grabbing, fingers flaring on the throw, the power rippling the fingers and, at rest, the gun raised to look at it). Over everything, the **final «serious PS2» pass** — film colour, soft glows, vignette and grain — as in Silent Hill 2, RE4, MGS3 live in their own scene, drawn after the world, so they never clip into walls; they sway with the stride, lag on turns, recoil when spraying, the left hand reaches to take and, **holding an element, keeps it alive on the palm like a plasmid** (sparks for metals, | `test_atomic` §6 — measured in the live scene: high is the starting graphics; ACES, sRGB, shadows, reflections, physical materials; **more than five glows** in scene; the protagonist with **two modelled eyes, five fingers**, a head above 3000 triangles and a figure above 15000; **no box** in the protagonist and the ten guardians; first person by default with the arms in their own scene, five fingers, the tool showing the compound held, spray droplets, a gaze that turns but not beyond a head; sodium sparking and chlorine as vapour on the left hand, the throw; brass and marble in the environments; the segmented health bar; a log with subtitles ending up in the Codex; the chemist moving forward facing where it goes and knees never backwards; the other way, PS1 at 480×360 with vertex lighting and 15-bit colour |

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| | | flame for fuels, vapour for gases, with a light illuminating the glove) and throws it; the gaze moves with the mouse (pointer lock) or by dragging a finger | |
| **ING-01** | One function must not have two implementations: the React/FastAPI stack must run THE SAME files as the application | `stack/api/worker/motore.mjs` — a Node process loads the repository modules with RDKit WebAssembly; no Python engine | `test_stack` — **15 checks**: compares the SHA-256 fingerprints declared by `GET /salute` with those of the files on disk, verifies the numbers match the browser's, and that no `.ts` / `.tsx` file contains a shift table |

## 2. AI agent requirements

| ID | Requirement | Implementation | Verification bench |
|---|---|---|---|
| **AI-01** | A model retired by the provider must not block the assistant | `modelloSuggeritoDaErrore()`, `bsi_modelli_ko` blacklist | `test_404` — 36 checks |
| **AI-02** | A provider unreachable from the browser must be remembered and demoted | `bsi_prov_ko`, 24 h TTL, `providerUtilizzabili()` | `test_ko` (35), `browser_ko` (37) |
| **AI-03** | A temporary overload must produce a retry, not an error | `erroreTemporaneo()`, `TENTATIVI_TEMPORANEO` | `test_503` — 40 checks |
| **AI-04** | An imposed wait longer than 10 s must switch provider if a ready one exists | `ATTESA_TROPPO_LUNGA_MS` | `test_attesalunga` — 14 checks |
| **AI-05** | The provider's token ceiling must be learned from the error, not hard-coded | `tettoDaErrore()`, `MARGINE_TETTO` | `test_tetto` — 32 checks |
| **AI-06** | Opaque reasoning signatures must be returned to the provider | capture and replay in `appendAgentTurn()` | `test_firma` — 31 checks |
| **AI-07** | Two overlapping submissions must not corrupt the history | `_turnoInCorso` in `send()` | `test_doppioinvio` — 7 checks |
| **AI-08** | The user must be able to measure which providers they can actually reach | `provaTuttiIFornitori()` | `browser_prova` — 29 checks |
| **AI-09** | Paid providers must not be offered as free ones | separation by `PROVIDERS[id].free` | `browser_prova` §D2 |
| **AI-10** | A documented route must exist to remove CORS and rate limits | `proxy/spectra-proxy.js` + in-app guide | `test_guidaproxy` (12), `browser_proxy` (15), `browser_proxyui` (17) |

## 3. Stability requirements

| ID | Requirement | Implementation | Verification bench |
|---|---|---|---|
| **STA-01** | A long session must accumulate neither DOM nodes nor timers | section lifecycle management | `audit_stabilita` §1 — 96 sections × 5 rounds |
| **STA-02** | Exhausting `localStorage` must not make the app unusable | guarded writes; persistent warning in the File Manager | `audit_quota` (10), `test_filemanager` (15) |
| **STA-03** | No rejected promise may remain unhandled | systematic `.catch()` | `audit_promesse` — 22 checks, healthy network and dead network |
| **STA-04** | Corrupted saved data must not prevent startup | `loadJSON()` with fallback | `audit_stabilita` §4 |
| **STA-05** | A slow network must not be treated as an absent one | `ATTESA_RETE_MS = 3500` in `sw.js` | `test_sw` §3 and §3-bis — measured 3,507 ms |
| **STA-06** | An update must not destroy work in progress | `BSI_SW_UPDATED` message, decision taken page-side | `test_sw` §4 |
| **STA-07** | Browsing molecules must not exhaust the WebGL contexts | reuse of the 3D viewer at all 5 creation points | `test_visore3d` — from 7 constructions down to 1 |
| **STA-08** | An unavailable IndexedDB store must be declared, not fatal | `dbStore()` over a promise; persistent warning | `test_filemanager` §2 |

## 4. Interface and accessibility requirements

| ID | Requirement | Implementation | Verification bench |
|---|---|---|---|
| **UI-01** | Every page must open without JavaScript errors | — | `audit_stabilita` §5 — 14 pages |
| **UI-02** | Charts must be sharp on high-density screens | `bsiNitido()` on the canvas contexts | `audit_grafici` — 40 canvases |

| ID | Requirement | Implementation | Verification bench |
|---|---|---|---|
| **UI-03** | The app must work at 390 px width, and content that does not fit must stay **reachable** | `auto-fit` grids, no fixed column, and wide tables inside a container with `overflow-x:auto` | `audit_mobile` — **two distinct measurements** across all 96 sections at 390 px: the document's `scrollWidth` (the horizontal bar the user sees) **and** elements wider than their section **with no scrollable ancestor**, that is content clipped and unreachable — a written column that cannot be read and gives no sign of existing. **Both ways**: a wide table inside a scrolling container is **not** reported, because it can be read by scrolling. `audit_stabilita` for errors in a phone viewport |
| **UI-04** | The user must be able to know which version they are running and force the update | "Updates" entry in the ✨ panel | `test_aggiorna` — 9 checks |
| **UI-05** | Data deletion must be selective and reversible in its choices | `bsiCancellaDati()` by group | `browser_reset` — 24 checks |
| **UI-08** | Surfaces that appear only after an action must also meet WCAG AA contrast | the six panels of the Cheminformatics section | `test_cheminfo` — the analysis is run, the panels opened in turn, **460 text elements measured, 0 defects**, with the WCAG formulas rewritten inside the bench |
| **UI-09** | Opening a section must not block the page | 3Dmol's first `render()` moved out of the click; the 296 synthesis figures **and the 263 drug cards** drawn in 8 ms slices | `test_fluidita` — **96 sections traversed**, median switch 16 ms, worst 87 ms, **0 sections above 100 ms** (there were 3 with the new drugs, at 159 ms); plus the opposite proofs: the WebGL canvas appears anyway, all 296 figures exist shortly after, and printing does not come out mute. For the viewer the claim is **structural, not temporal**: immediately after the click handler returns, the WebGL context must not exist; shortly after it must — because a millisecond threshold was measuring SwiftShader rather than the code (14…192 ms on identical code) |
| **UI-10** | The interface must switch among many languages, declaring how much each is translated, and carrying the writing direction with it | `bsi-lingue.js` with **fourteen languages**, a searchable selector, `dir="rtl"` for Arabic | `test_lingue` — **170 skeleton elements out of 170 in each of the thirteen languages**, coverage computed by applying the dictionary and counting; writing direction tested in both directions; navigation categories and the search placeholder included |

| ID | Requirement | Implementation | Verification bench |
|---|---|---|---|
| **UI-11** | The molecule must show how it holds together, and the viewer must choose the angle | `bsi-mol3d.js`: formation from dust with staggered arrival, and atom selection by click | `test_mol3d` — the canvas **pixels** at 130 ms and 2.3 s, the arrival order (skeleton before hydrogens) measured on each atom's progress, and the selection of 1, 2, 3 and 4 atoms |
| **UI-12** | Whoever has a file must be able to open it: a picker listing extensions does not narrow the choice on a phone, it **hides** it | `bsi-spettrolettore.js` — a single `<input type="file">` **with no `accept`**, a single button, and `apriQualunque()` routing by type, by the extension the format declares and **by content**, announcing the decision in `#bsiSP-deciso` | `test_spettrolettore` §7 — one visible button, an **empty** `accept` (a non-empty `accept` here *is* the defect), and routing verified on PDF, `.docx`, `.txt`, `.jdx`, two `.csv` columns and an image; **both ways**, a binary disguised as `.txt` is not poured out as text but declared binary |
| **UI-13** | The version shown in the header must be the one actually running | `index.html` — the badge is filled from `window.BSI_APP_VERSION`, no longer a hand-written string | `test_spettrolettore` §8 — the badge text is compared with `BSI_APP_VERSION` measured in the running page. It was needed: for about eighty versions the header read `v13l` when the application was already past version 190, and no check looked at it |

## 5. Security and privacy requirements

| ID | Requirement | Implementation | Verification bench |
|---|---|---|---|
| **SEC-01** | No API key may be present in the repository | keys only in `localStorage` or in the Worker's secrets | `tools/verifica-sicurezza.js` — 381 tracked files, 8 credential shapes |
| **SEC-02** | No personal data may leave the device without an explicit action | local-first architecture, telemetry disabled | `audit_rete` — **direct verification**: a canary value seeded into 71 stores of the user's data, the application used across 96 sections on 6 pages, and the URL, headers and body of every request inspected. Plus `verifica-sicurezza` on the two exit mechanisms |
| **SEC-03** | Passwords must not appear in clear text in the source | SHA-256 in `file_manager.html` | `tools/verifica-sicurezza.js` |
| **SEC-07** | No credential may EVER have entered the repository, not even in a commit later fixed | no key has ever been committed; keys live in `localStorage` or in the Worker's secrets | `audit_storia` — **1,551 distinct versions of text files across 390 commits**, 9 patterns tested in both directions; fails on a shallow clone, because it would measure less surface |
| **SEC-04** | No conflict marker may reach publication | — | `verifica_guida` §12 — 58 text files |

> **SEC-03 — declared limitation.** GitHub Pages serves static files: any page-side access control can be bypassed by whoever reads the source. The File Manager's protection is a deterrent, **not** a security control, and the documentation says so. Furthermore the password remained in clear text in the git history until its removal, and taking a secret out of the files does not take it out of the history: `git log -p` hands it to anyone. The only effective remedy was to change it, and that **has been done** at version `bsi-v188`. The old one remains in the history and no longer opens anything.

## 6. Requirements not yet verified automatically

Declared for completeness. They are covered by documentary inspection or by manual testing, and their automation is planned.

| ID | Requirement | Current coverage |
|---|---|---|
| **UI-06** | Full WCAG 2.1 AA conformance | the mechanical part is automated across **13 pages and 96 sections** (`tools/verifica-accessibilita.js`, 38,407 text elements: it was 19,751 before text over gradients entered the measurement, and 37,409 before the «Spectrum reader» section). **Of the 1,069 contrast defects that emerged, none remain: 0 measured** over those same 96 sections, and the value is recorded as a baseline the bench defends. Unlabelled fields are **zero**: see the box in `docs/09` §4. What remains outside is text inside SVGs and text over a real background image, counted on every run (D-09) |
| **PERF-01** | First-paint time on a low-end device | manual cross-device testing (`docs/02` §4) |
| **SCI-11** | Structures of 2 entries that are not single molecules (it was 6) | **not representable**: Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. The other four were closed by taking the structure from ChEMBL, see `docs/06` §2.4 |
| **PERF-02** | Code coverage of the verification benches | **partially measured**: 49.89 % of statements over the widest path (`audit_copertura`). Whole-battery coverage and branch coverage remain unmeasured; see §8 |
| **UI-07** | Behaviour on Firefox and WebKit | **not verified** — the battery runs on Chromium only; see §8 |

## 7. Overall coverage

| Category | Requirements | Verified by a bench | Coverage |
|---|---:|---:|---:|
| Scientific (SCI-01…37) | 36 | 36 | 100 % |
| AI agent (AI-01…10) | 10 | 10 | 100 % |
| Stability (STA-01…08) | 8 | 8 | 100 % |

| Category | Requirements | Verified by a bench | Coverage |
|---|---|---|---|
| Interface (UI-01…19) | 17 | 17 | 100 % |
| Security (SEC-01…07) | 5 | 5 | 100 % |
| Architecture (ING-01…02) | 2 | 2 | 100 % |
| **Automated total** | **76** | **76** | **100 %** |
| Not automated (§6) | 5 | 0 | 0 % |
| **Declared total** | **81** | **76** | **94 %** |

Coverage is computed over the requirements **declared in this document** and must not be confused with code coverage: it measures how many requirements have a bench verifying them, not how many lines are executed during the tests.

> **Why two totals.** Reporting only the 100 % of automated requirements would be true and misleading at once: it is 100 % of what was *chosen* to be automated. The number that matters to an assessor is the second one — **54 requirements verified out of 59 declared** — and the five that are missing are listed by name in §6, not summarised into a percentage.
>
> This table is checked by `tools/verifica-documenti.js`, which counts the identifiers actually present in the document and fails if the totals written here do not match. It is a check born of a real error: the totals said 36 over 9 scientific categories while the identifiers defined were 10, and `SCI-10` appeared twice — once as a verified requirement and once as an unverified one.

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## 8. Declared coverage gaps

These are not defects: they are things that have **not been measured**, and that therefore cannot be asserted.

| Gap | What it entails | Why it is so |
|---|---|---|
| **Partial code coverage** | Measured: **49.89 %** of statements over the widest path a bench walks. It is not the coverage of the whole battery, and it is statements, not branches | Chromium's profiler collects it inside the engine, with no build and without rewriting the source: the obstacle belonged to the tool (c8, istanbul), not to the problem. Value recorded; `audit_copertura` fails if it falls |
| **Chromium only** | Behaviour on Firefox and WebKit is verified by hand, not by a bench | The battery uses `playwright-core`, which downloads a single engine. In the verification environment the CDN for the other engines answers **403** to the network policy: they cannot be installed there |
| **No visual regression** | An unintended graphical change would not be caught | Spectra are deterministic and comparable byte for byte (SCI-05): the comparison exists on the traces, not on the whole page |

| Gap | What it entails | Why it is so |
|---|---|---|
| **2 entries without a structure** | Two entries out of 178 have no structure to show: Ivermectin is a mixture of homologues, Coartem a combination of two active ingredients. It was six | See `06-Scientific-Accuracy-Data-Provenance.md` §2.4 |
| **Performance on slow devices** | First-paint time is not measured on low-end hardware | It requires physical devices; the test is manual |

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_Document updated to version `bsi-v216`._